



These are
regular contacts,
not the Zoom lens

Zoom lens

Scientists have created contact lenses that can zoom in when you blink twice. Sounds like the stuff of science fiction movies, doesn't it? But it's apparently real, scientists at the University of California San Diego have actually gone ahead and manufactured such a lens.

The lens is controlled by eye movements, one of which includes being able to zoom in on stuff if you blink twice.

How does it work? Scientists measured the electrooculographic signals generated by the eyes when they make specific movements. This could be anything, right from looking in different directions to blinking, or blinking twice in this particular case. They then created a soft biometric lens that responds to these signals. The lens itself is able to change its focal length depending on the signal being generated.

Since the lens works on the electric signals created by the movement of eyes, they'll work even if the user can't actually see.

Arque Robotic Tail

Of course, who else but Japan would come up with the idea of a wearable tail. No, not a prop tail, an actual working tail. A tail that actually helps improve balance and agility, you know, the way a tail works for just any animal that actually has one.

Dubbed Arque, the tail is pretty much designed to be an extension of your body, much like your limbs. The creators of the tail at Keio University in Japan drew inspiration from monkeys, who use their tails in a myriad of ways, from swinging between trees to maintain their balance and shifting their centre of gravity.



The tail, however, doesn't look like what you'd imagine a tail to look like, it was created by interlocking multiple plastic vertebrae. This way, not only is it an extension to your spine, but it also allows for easy modification, should you need to add or remove blocks as per the user. The tail is powered with compressed air, and uses artificial muscles to contract or expand to make the tail move.

DJI FPV Transmission System

FPV Drones, or First-Person View Drones have been around for a while, however, they've had terrible quality when it came to controlling them and to the video quality itself. Then there's the added issues of poor connection and issues with the video feed. Apparently, DJI claims to have addressed these issues and then some with their new Digital First Person Viewing Transmission System.

The FPV system comes equipped with four components, which include a streaming module, a 4MP camera with a 1/2.3-inch

CMOS module and a 150 degree FoV, a remote control, and goggles, which have 1440x810, 120Hz displays in them.

The streaming module has a range of 4km, and a max latency of 28ms, or milliseconds. The module captures at 720P with 120 frames per second,



and those frame rates would definitely help improve the FPV experience.

The goggles can record and store video at 1080p and 60 frames to the in-built microSD cards that are present in them.

The system uses clever techniques to ensure that video quality doesn't suffer while in use. One such technique is lowering the quality of the video near your peripheral vision and focusing attention and quality on the centre of the image.

The system also comes with three different video modes, with one of them optimised for drone racing.